

INTRODUCTION

In all engineering construction, the component parts of a structure or a machine must be assigned definite physical sizes. Such parts must be properly proportioned to resist the actual or probable forces that may be imposed upon them.

- Thus for example the walls of a pressure vessel (Such as boiler) must be of adequate strength to withstand the internal pressure.
- The floors of a building must be sufficiently strong for their intended purpose.
- The shaft of a Machine must be of adequate size to carry the required torque.
- A wing of an airplane must safely withstand the aerodynamic loads that may come upon it during take off, flight and landing.
- Likewise the parts of a composite structure must be rigid enough so as not to deflect excessively when in operation under the imposed loads.

The ability to determine the minimum load that an element of a structure can carry before failure happens is of great importance. In engineering practice such requirements must be met with the minimum expenditure of a given material.

- The subject strength of materials' involves analytical methods for determining the strength, stiffness and stability of various load carrying materials.
- This subject is very important to be studied by many braches of engineering. It is important for *civil engineers* in the design of bridges and buildings, for *nuclear engineers* in the design of reactor components, for *mechanical engineers* for the design of machineries and pressure vessels, for *metallurgical engineers*, who need fundamental concepts of this subject in order to understand how to improve existing materials further.